



Lab

The next few weeks

This Week

QR Factorization

SVD (Part I)

QR Factorization

Next Week

SVD (Part II, recorded)

Sample exam problems?

SVD

Week 12

Eigenvectors (Part I)

Eigenvectors (Part II)

Eigenvectors

Sample exam
problem review

Lecture 20: Singular Value Decomposition (first part)

Instructors:

Christoph Lohrmann (christophl@ru.is)

Álfur Birkir Bjarnason (alfurb@ru.is)

[Based on slides for „Decoding the Matrix“]



WEEK

TEN



Agenda

Chapter 11

Lecture 20:

- Low rank matrix
- Matrix norm
- Trolley location problem
- Best rank-one approximation
- Singular value and Singular vectors





Lecture 19 - Summary

Previous

orthogonalization

Orthogonal complement

Previous
Tools

QR factorization

orthogonalize

Solving $Ax = b$,
A square matrix

Solving $Ax = b$,
A rectangular matrix

Least square regression

High-
Dimensional
Fire Engine
Lemma



Lecture 20 – goal: Introduce low-rank matrix



PÓLITÍSK FULLYRÐING

Íslenskt samfélag einkennist af réttlæti, sanngirni og jöfnum tækifærum.

☐ Mjög mikilvæg fullyrðing



Mjög ósammála



Nokkuð ósammála



Nokkuð sammála



Mjög sammála

[Sleppa fullyrðingu](#)

Alþingiskosningar 2024

Svona dreifast svör allra frambjóðenda í kosningaprófi RÚV

Hátt í 700 frambjóðendur svöruðu kosningaprófi RÚV. Þeir hafa jafn misjafnar skoðanir og þeir eru margir. Svör frambjóðenda Lýðræðisflokksins eru að jafnaði lengst frá svörum eigin flokks.

Birgir Þór Harðarson og Émile Nadeau
29. nóvember 2024 kl. 06:48, uppfært kl. 10:54

● Frambjóðendur ■ Flokkur





Silly compression

Matrix $\mathbf{A}$,
 $\mathbf{m} \times \mathbf{n} = 625 \times 1024$
625k elements



Low rank matrix $\tilde{\mathbf{A}}$, rank-50
 $\mathbf{r}(\mathbf{m} + \mathbf{n}) = 50(625 + 1024)$
82k elements





Frobenius norm for matrices

Proposition: Squared Frobenius norm of a matrix is the sum of the squared norms of its rows ... or of its columns.

$$\left\| \begin{bmatrix} \mathbf{v}_1 & \cdots & \mathbf{v}_n \end{bmatrix} \right\|_F^2 = \|\mathbf{v}_1\|^2 + \cdots + \|\mathbf{v}_n\|^2$$

The trolley line compression problem



Given the locations of m houses $\mathbf{a}_1, \dots, \mathbf{a}_m$, we must choose where to run a trolley line.

The trolley line must go through downtown (origin) and must be a straight line.

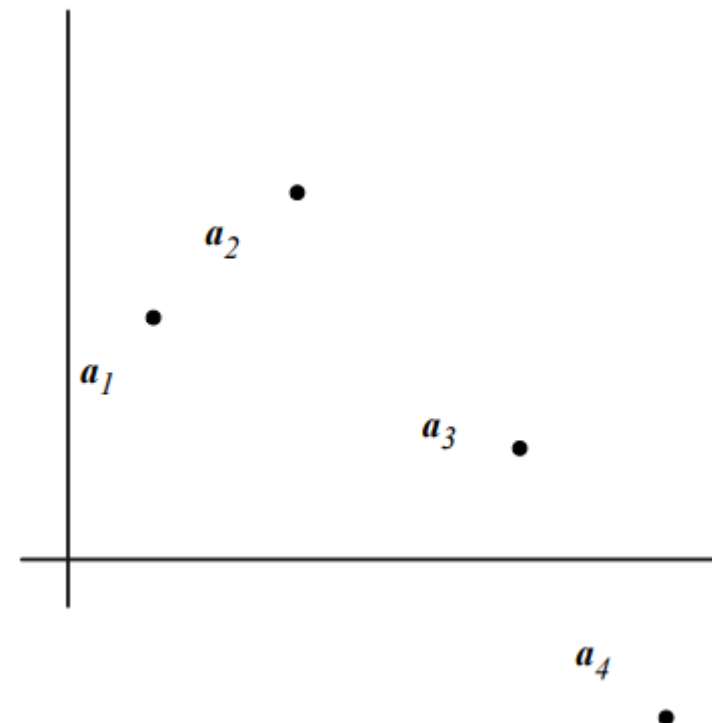
The goal is to locate the trolley line so that it is as close as possible to the m houses.

Specify line by unit-norm vector $\mathbf{v}$: line is $\text{Span}\{\mathbf{v}\}$.

In measuring objective, how to combine individual objectives?

As in least squares, we minimize the 2-norm of the vector $[d_1, \dots, d_m]$ of distances.

Equivalent to minimizing the square of the 2-norm of this vector, i.e. $d^2 + \dots + d^2$.





Solution to the trolley line location problem

For each vector $\mathbf{a}_i$, write $\mathbf{a}_i = \mathbf{a}_i^{\parallel \mathbf{v}} + \mathbf{a}_i^{\perp \mathbf{v}}$ where $\mathbf{a}_i^{\parallel \mathbf{v}}$ is the projection of $\mathbf{a}_i$ along $\mathbf{v}$ and $\mathbf{a}_i^{\perp \mathbf{v}}$ is the projection orthogonal to $\mathbf{v}$.



Solution to the trolley line location problem

For each vector $\mathbf{a}_i$, write $\mathbf{a}_i = \mathbf{a}_i^{\parallel \mathbf{v}} + \mathbf{a}_i^{\perp \mathbf{v}}$ where $\mathbf{a}_i^{\parallel \mathbf{v}}$ is the projection of $\mathbf{a}_i$ along $\mathbf{v}$ and $\mathbf{a}_i^{\perp \mathbf{v}}$ is the projection orthogonal to $\mathbf{v}$.



Solution to the trolley line location problem continued

$$\sum_i (\text{distance from } \mathbf{a}_i \text{ to Span } \{\mathbf{v}\})^2 = \|A\|_F^2 - \|A\mathbf{v}\|^2$$

Therefore the best vector $\mathbf{v}$ is a unit vector that maximizes $\|A\mathbf{v}\|^2$ (equivalently, maximizes $\|A\mathbf{v}\|$).



The trolley line location example:

Example: Let $\mathbf{A}$ be a matrix and $\mathbf{v}_1, \mathbf{v}_2$, two unit-norm vectors, show that $\mathbf{v}_1$ is a better solution to the trolley line problem than $\mathbf{v}_2$:

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ -1 & 2 \end{bmatrix}$$

$$\mathbf{v}_1^T = \begin{bmatrix} 0 & 1 \end{bmatrix}$$

$$\mathbf{v}_2^T = \begin{bmatrix} 1 & 0 \end{bmatrix}$$



Solution to the trolley line location problem

$$\sum_i (\text{distance from } \mathbf{a}_i \text{ to Span } \{\mathbf{v}\})^2 = \|A\|_F^2 - \|A\mathbf{v}\|^2$$

Therefore the best vector $\mathbf{v}$ is a unit vector that maximizes $\|A\mathbf{v}\|^2$ (equivalently, maximizes $\|A\mathbf{v}\|$).

```
def trolley_line_location(A):  
    Given a matrix  $A$ , find the vector  $\mathbf{v}_1$   
    minimizing  $\sum_i (\text{distance from row } i \text{ of } A \text{ to Span } \{\mathbf{v}_1\})^2$   
     $\mathbf{v}_1 = \arg \max \{\|A\mathbf{v}\| : \|\mathbf{v}\| = 1\}$   
     $\sigma_1 = \|A\mathbf{v}_1\|$   
    return  $\mathbf{v}_1$ 
```



Theorem:

Theorem: Let A be an $m \times n$ matrix over $\mathbb{R}$ with rows $\mathbf{a}_1, \dots, \mathbf{a}_m$. Let $\mathbf{v}_1$ be the first right singular vector of A . Then $\text{Span} \{\mathbf{v}_1\}$ is the one-dimensional vector space $\mathcal{V}$ that minimizes

$$(\text{distance from } \mathbf{a}_1 \text{ to } \mathcal{V})^2 + \dots + (\text{distance from } \mathbf{a}_m \text{ to } \mathcal{V})^2$$

How close is the closest vector space to the rows of A ?

Lemma: The minimum sum of squared distances is $\|A\|_F^2 - \sigma_1^2$.



Example:

Example: Let $\mathbf{A}$ be a matrix and $\mathbf{v}_1$, unit-norm vector and solution to the trolley line problem, check the validity of the previous lemma:

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ -1 & 2 \end{bmatrix}$$

$$\mathbf{v}_1^T = \begin{bmatrix} 0 & 1 \end{bmatrix}$$

$$\sigma_1 = \sqrt{8}$$



Best rank-one approximation to a matrix:

A rank-one matrix is a matrix whose row space is one-dimensional.

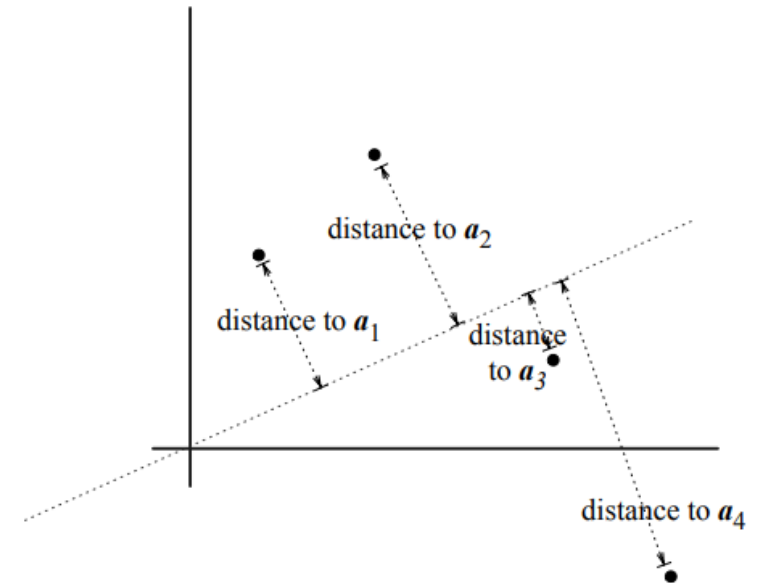
All rows must lie in $\text{Span}\{\mathbf{v}\}$ for some vector $\mathbf{v}$.

That is, every row is a scalar multiple of $\mathbf{v}$.

Theorem 11.2.9: The rank-one matrix $\tilde{A}$ that minimizes $\|A - \tilde{A}\|_F$ is

$$\tilde{A} = \begin{bmatrix} \text{vector in Span}\{\mathbf{v}_1\} \text{ closest to } \mathbf{a}_1 \\ \vdots \\ \text{vector in Span}\{\mathbf{v}_1\} \text{ closest to } \mathbf{a}_m \end{bmatrix}$$

and, for this choice, $\|A - \tilde{A}\|_F^2 = \|A\|_F^2 - \sigma_1^2$.





Best rank-one approximation

Definition: The *first left singular vector* of A is defined to be the vector $\mathbf{u}_1$ such that $\sigma_1 \mathbf{u}_1 = A\mathbf{v}_1$, where σ_1 and $\mathbf{v}_1$ are, respectively, the first singular value and the first right singular vector.

Theorem: The best rank-one approximation to A is $\sigma_1 \mathbf{u}_1 \mathbf{v}_1^T$ where σ_1 is the first singular value, $\mathbf{u}_1$ is the first left singular vector, and $\mathbf{v}_1$ is the first right singular vector of A .



Application to voting data:

Let $\mathbf{a}_1, \dots, \mathbf{a}_{100}$ be the voting records for US Senators.

Snowe	0.106605199	moderate Republican from Maine
Lincoln	0.106694552	moderate Republican from Rhode Island
Collins	0.107039376	moderate Republican from Maine
Crapo	0.107259689	not so moderate Republican from Idaho
Vitter	0.108031374	not so moderate Republican from Louisiana

Look at projection along $\mathbf{v}$ of each of these vectors.





The closest one-dimensional affine space

In *trolley-line problem*, line must go through origin:
closest one-dimensional *vector space*.

Perhaps line *not* through origin is much closer.

An arbitrary line (one not necessarily passing through the origin) is a one-dimensional *affine* space.

Given points $\mathbf{a}_1, \dots, \mathbf{a}_m$,



Politics revisited:

We center the voting data, and find the closest one-dimensional vector space $\text{Span}\{\mathbf{v}_1\}$.

Now projection along $\mathbf{v}$ gives better spread.



Which of the senators to the left of the origin are Republican?

```
>>> {r for r in senators if is_neg[r] and is_Repub[r]}
{'Collins', 'Snowe', 'Chafee'}
```

Similarly, only three of the senators to the right of the origin are Democrat.



Lecture 20 - Recap

Previous

QR factorization

Tool

Trolley solution

Previous

Pythagorean theorem

Concept & Definition

Low rank matrix

Frobenius norm

Trolley line problem

Best rank-one approximation

First Singular value, Right Singular vector, Left Singular vector

Data centering



Thank You For Your Attention!





Applied exercise (1)

Problem: Let A be a matrix and Q a column-orthogonal matrix, give a numerical example that counter this statement: $\|AQ\|_F = \|A\|_F$



Applied exercise (2)

Problem: Let be A a square matrix of dimension 4×4 .

For the right singular vector vector v_1 , calculate the first singular value σ_1 , the left singular

vector u_1 and the Frobenius norm $\|A - \tilde{A}\|$

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$v_1^T = \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix}$$